CANADIAN NATIONAL RAILWAY

TRANSPORTATION OF DANGEROUS GOODS

Effective 0001 on

May 1, 2023

TRANSPORTATION DG

General

The transportation of dangerous goods by rail is governed by the Transportation of Dangerous Goods Act and Regulations. Violation of these regulations can result in fines or conviction.

Crews destined to the U.S. must have a **copy** of the NORTH AMERICAN EMERGENCY RESPONSE GUIDEBOOK accessible.

Crew members must carry a valid Dangerous Goods Training Certificate while on duty.

Electronic Documentation may be used for Train Journal (WOPRT) provided employees are also carrying the Transportation Canada Equivalency Certificate SR13315 (Ren.1) on the same device

Where instructions indicate notations and/or initials must be added to the Train Journal (WOPRT), these modifications will be made on the E-Train Journal Form when using an E-Train Journal. Cars that are set out will be indicated on the E-Train Journal.

Canadian and US Dangerous Goods Differences

DG classification, including hazard classes, differ between US and Canada. As such, any commodity that travels under NAID instead of UNID would not be a DG in Canada or count towards Key Train or Higher Risk Key Train.

Example: The commodity is moving as NA1993 which is not regulated in Canada but if it moving as UN1993 it would be regulated in Canada.

Section 1.0 Inspection

Before lifting placarded dangerous goods cars from shippers tracks or interchange tracks, in addition to the inspection requirements of G.O.I section 5 item “Inspection of standing equipment”, where applicable, crews must visually inspect (from a position on the ground) to verify:

- All valves, lids and handles are closed, caps installed and there are no signs of leakage.
- All required markings are present.
- Any signs of tampering, such as suspicious items or items that do not belong, the presence of an “Improvised Explosive Device” (IED), and other signs that the security of the car may have been compromised.

Note: Where an indication of tampering or a foreign object is found, take the following actions:

1) Do not accept or move the rail car.
2) Immediately move yourself and others to a safe location away from the rail car before using radios and cell phones to make notifications.
3) For cars at a customer’s facility, immediately contact local plant personnel. If local plant personnel are not available or cannot explain what you see, immediately contact your transportation supervisor.
4) For cars on interchange tracks or in the yard, immediately contact the yardmaster or transportation supervisor.

Section 2.0 Placarding

(a) Before lifting dangerous goods cars from shippers or interchange tracks, crews must have verification, or inspect to ensure that the required number of placards are affixed to the car, and with the exception of mixed loads (DANGER placard), the placards indicate the class of the commodity as shown on the shipping documents.

Exception:

Tank cars of Molten Sulfur UN 2448 can be moved without placards affixed to the car, as long as the identification number UN 2448 and the words “MOLTEN SULFUR” appear on each side of the tank car.

Exceptions 2:

Cars handling Class 9 commodities with an Identification Number NA as indicated on the shipping document, can be moved in Canada without placards. These cars, if going to or coming from the United States, may display the identification number in an orange panel or in a white diamond-shaped placard on all four sides of the rail car.

(b) Placards will be displayed on both sides and both ends of a rail car, including each compartment of a compartmentized tank.

Exception: Placards will be clearly visible on both sides of Containers, trailers and other large means of containment containing dangerous goods, being transported on rail cars.

(c) Missing or damaged placards must be promptly reported to those who will arrange for replacement. Dangerous Goods cars missing placards must not be lifted.

Section 3.0 Documentation

(a)

i When lifting from shippers tracks or interchange tracks, and while enroute, crews are responsible to be in possession of the shipping document for each loaded or residue dangerous goods car.

ii An SRS waybill (follower lines on WOPRT or Train Lists) is an acceptable shipping document for any loaded or residue dangerous goods car/intermodal traffic.

iii If the SRS waybill information for the WOPRT generated shipping document does not meet all the TDG requirements, the shipment will require a separate shipping document to accompany the car.

The WOPRT will list the cars requiring separate documentation and in addition, each car will have a follower line stating **DANGEROUS GOODS SHIPPING DOCUMENTS - REQUIRED**.

iv The shipping document may not indicate whether the car contains a “Special Dangerous Commodity”. When a loaded placarded car is lifted enroute, and a consist or other means is not available to indicate that the car is not a “Special Dangerous Commodity”, such car must be handled applying “Special Dangerous Commodity” speed restrictions and inspection instructions until such time as it can be verified otherwise.

(b)

i Possession of documentation by the crew is not applicable when switching in make up yards, or where cars are being switched onto trains or into classification tracks provided the required documentation is available at the responsible railway office.

ii Where Dangerous goods cars are to be placed/ pulled on/from a consignee's track within a yard or a terminal area, the crew must be in possession of the shipping document for each dangerous goods car. Switch lists can be generated containing the required shipping document if a separate shipping document is not provided.

(c) Unless relieved of the responsibility, crews are responsible for leaving such documentation at designated locations when setting out or placing cars.

Note: Crews may be relieved of this responsibility:

- when setting out cars, (e.g bad order or set off) provided that the shipping document is available at the responsible railway office;
- when placing cars on a customers track, the shipping document has been delivered to the consignee or the interchange railway.

(d) Whenever a car containing dangerous goods is set off, the RTC will instruct a crew member as to the disposition of the documentation.

(e) After leaving a terminal, if the required documentation is lost, or it is discovered that a dangerous goods car is in their movement without the required documentation, the RTC must be notified. The crew will be provided details of the commodity in the form of a “Radio Waybill” with the applicable items shown below copied by a crew member. This information will be retained until the proper required documentation is provided to the crew.

1. INITIAL/NUMBER
2. ORIGIN
3. SHIPPER
4. SHIPPER STREET ADDRESS
5. DESTINATION
6. CONSIGNEE
7. CONSIGNEE STREET ADDRESS
8. CAR/PACKAGE TYPE
9. RESIDUE LAST CONTAINED
10. UN/NA NUMBER
11. PROPER SHIPPING NAME
12. CLASS/SUBCLASS
13. COMPATIBILITY GROUP
14. PACKING GROUP
15. POISON INHALATION HAZARD (INCLUDING ZONE A/B/C/D)
16. QUANTITY AND UNIT OF MEASURE
17. REPORTABLE QUANTITY
18. EMERGENCY 24HR. NO.
19. ERAP NUMBER
20. ERAP PHONE
21. WB DATE
22. SPECIAL COMMODITY
23. MARINE POLLUTANT
24. POISON/TOXIN
25. ELS PERMIT AND EXPIRY DATE
26. DOT EXEMPTION NUMBER
27. NET EXPLOSIVE QUANTITY
28. FLASH POINT
29 EMERGENCY CONTROL TEMPERATURE
30. SPECIAL INSTRUCTIONS
31. DOT 113 - DO NOT HUMP OR CUT OFF CAR IN MOTION
32. FUMIGATED UNIT

(f) Railway Security Sensitive Material (RSSM)

The following instructions apply to carloads which are considered " Railway Security Sensitive Material" and may or will operate through, or be lifted or placed at a customer siding, within a High Threat Urban Area (HTUA) within the U.S.

- FOLLOWER LINE: These carloads will be identified on all train lists with a follower line that indicates "RSSM".
- CHAIN OF CUSTODY: A documented chain of custody requirement is now mandated. These identified carloads (RSSM) must be "attended" on hand-off and/or receipt with a customer or other railway and a transfer of custody recorded.

  That is, when lifting or placing cars on a customer siding, delivering or receiving to/from a foreign railway directly or in interchange, with RSSM carloads present, the involved crew at the time must identify the receiver or delivering party, and both parties must sign-off and keep record of the transfer of custody of the carloads. The one exception to this requirement occurs where the final destination for the car is to a customer outside of a HTUA, in which case attendance and a chain of custody report is not required.

  The following is a list of HTUA locations: Buffalo, Detroit, Toledo, Pittsburgh, Chicago, Milwaukee, Minneapolis, Omaha, St. Louis, Memphis, Baton Rouge and New Orleans.

As CN is not always made aware of the final destination of cars interchanged, all RSSM must be considered as being destined to HTUA and the chain of custody instructions must be applied.

- DOCUMENTATION: All train lists with RSSM carloads present will have a pre-printed form identified as the "RSSM - Transfer of Custody Supplemental Work Order" which will list all RSSM carloads and provide a form outlining date-timelocation- number of RSSM carloads-delivering employee and accepting employee. The header or summary page of all train journals (WOPRT) will have a field identifying if the movement has any RSSM carloads present, providing a total number. In addition the presence of a follower line will identify each individual carload on any train list. The conductor of the movement handling or receiving RSSM identified carloads will be responsible to ensure proper compliance and record as required by the regulation is achieved.

  It will be kept on record for 60 days.

  The "Transfer of Custody Record" must be delivered with the train journal, at the normal locations where train journals are now turned in.

Section 4.0 Train Consists

(a) Crews shall have in their possession a document indicating the position of each placarded car in their train. When the position is changed, (e.g. cars lifted or set off) or a placarded car is placed in the train, the document must be kept up to date and modified to indicate the change. Modifications must be made on the E-Train Journal Form when the E-Train Journal is used or an updated version of the E-Train Journal must be saved on the device when available.

(b) Any lifts or set-offs must be identified on the original document. In the case of an enroute lift, where a list of cars is provided, it must be verified for accuracy and included with the original document which must have a written insertion mark as indication of where the cars were placed in the train or transfer. The Conductor must sign the document as indication of correctness and as required by regulation or for use by emergency personel. A train consist, switch list, or other prepared document may be used to meet this requirement.

(c) When lifting, and a train list, switch list, or other prepared document is not provided for the car(s) lifted, the following must be recorded on the train's existing list:

- The car initial and number;
- UN number (for a car containing one commodity) or the words " d a n g e r o u s goods" for a car which contains more than one commodity. (e.g. mixed load)

(d) Carloads in the Toxic Inhalation Hazard (TIH) category, are restricted to a maximum speed of 50 MPH while in transit within the U.S. A follower line on all train lists will include the information: '50 MPH in USA'.

Section 5.0 Switching

(Note: these restrictions are in addition to those restrictions contained in CROR Rules 112 to 116)

(a) Humping operations will be governed by the HPCS procedures outlined in Terminal Manuals.

(b) Any impact suspected of being in excess of 6 MPH, with or onto a dangerous goods car must be promptly reported to the appropriate supervisor for furtherance.

(c) Cars in class Explosive 1.1, 1.2 or Poison gas 2.3 must not be placed under a bridge or overpass, nor in or alongside a passenger station.

(d) AAR 204 and DOT 113 specification tank cars will have a follower line indicating these cars must NOT be

- uncoupled while in motion;
- coupled into with more force than necessary to complete the coupling; or
- struck by any railway vehicle moving under it's own motion.

These cars are stenciled accordingly.

(e) Load and empty Poison or Toxic Inhalation Hazard (PIH or TIH) cars must not be let run free (except for humping operations). Cars must be shoved to rest.

PIH and TIH cars are identified as follow:

- a follower line on switching list and journal which read POISON INHALATION HAZARD **MUST BE SHOVED TO REST**
- cars are stenciled with ''INHALATION HAZARD'' on both sides.

Section 6.0 Marshalling

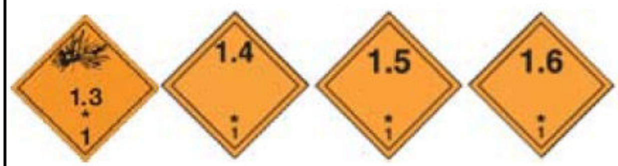
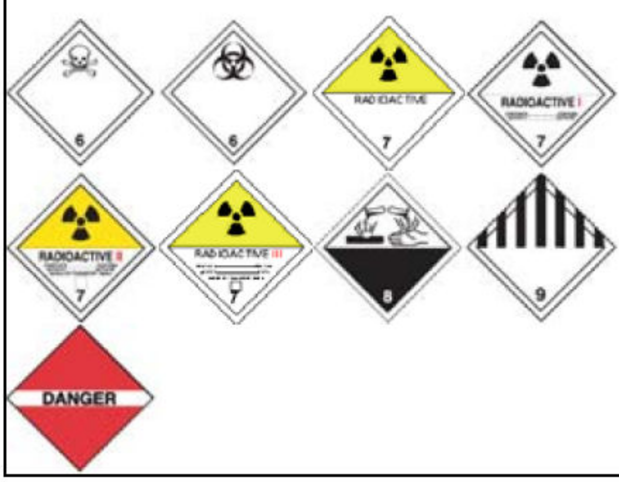
Note1: Any movement which will enter the U.S. must be marshalled in accordance with U.S. Regulations (49 CFR). U.S. marshalling requirements are provided in the U.S. Position in Train Chart (item 10.0) at the end of this section.

Note2: Unless relieved by follower line on WOPRT/ Switch lists (e.g. no marshalling required UN/NA 2448 Molten Sulphur), these marshalling restrictions apply to all placarded cars (loads and residues) on movements which will exceed 15 MPH.

(a) **General Restrictions** - Any placarded dangerous goods car must not be marshalled next to:

- an operating locomotive; (unless all cars in the train have a placard)
- any occupied car; (unless all other cars in the train have a placard)
- a car equipped with a heating or cooling device or has a source of ignition;
- any open top car;
    - i when the lading protrudes beyond the car and may shift during transport, or
    - ii when the lading is higher than the top of the car and may shift during transport.

(b) **Marshalling Chart -** Placarded dangerous goods cars are subject to the following marshalling group restrictions in addition to general restrictions.

<table>
<tr><th rowspan="2">DANGEROUS GOODS
MARSHALLING GROUP</th><th colspan="3">Must not be next to</th></tr>
<tr><th>Group A</th><th>Group B</th><th>Group C</th></tr>
<tr><td>Group A: Explosives 1.1 and 1.2
</td><td></td><td>X</td><td>X(1)</td></tr>
<tr><td>Group B: (If handled, see list below)</td><td>X</td><td>X(2)</td><td>X</td></tr>
<tr><td>Group C: Explosives 1.3 to 1.6

Classes 2, 3, 4 and 5
</td><td>X(1)</td><td>X</td><td></td></tr>
<tr><td>Groupe D: Classes 6, 7, 8, 9 and Mixed Loads
</td><td colspan="3">Only General Restrictions (a) as above.</td></tr>
</table>

Marshalling Chart Footnotes:

(1) not applicable to explosives Classes 1.3 to 1.6

(2) not applicable if the next car has the same UN number.

List of Group B Dangerous Goods	
UN 1008, CLASS 2.3	UN 1026, CLASS 2.3
UN 1051, CLASS 6.1	UN 1076, CLASS 2.3
UN 1067, CLASS 2.3	UN 1589, CLASS 2.3
UN 1614, CLASS 6.1	UN 1660, CLASS 2.3
UN 1911, CLASS 2.3	UN 1975, CLASS 2.3
UN 2204, CLASS 2.3	UN 2188, CLASS 2.3
UN 2199, CLASS 2.3	UN 3294, CLASS 6.1

(c) **Marshalling of plain bearing cars**

When there are plain bearings ahead, loaded dangerous goods cars must be marshalled:

- within the first 2000 feet on trains 4000 ft. or less; or,
- must not be in the last 2000 ft. on trains over 4000 ft.

(d) **Marshalling of placarded trailers/containers on intermodal rail cars.**

Containers/Trailers placarded as “Explosives 1.1, 1.2, or “ Radioactive class 7" must be loaded/ marshalled on an intermodal railcar so it will NOT be the first platform/car next to the locomotive, and in addition, explosives 1.1 and 1.2 must not be loaded/marshalled next to a container/trailer/ car which has a mechanical heating/cooling device.

For trains destined to the U.S., Explosives 1.1 and 1.2” must be loaded/marshalled on an intermodal railcar so it will NOT be nearer than the sixth platform/car next to the locomotive.

(e) **Marshalling of Distributed Braking cars / container (DBC)**

These are cars equipped with diesel powered air compressor and fuel tank which must not be marshalled next to:

- any occupied car; (unless all other cars in the train have a placard)
- the following classes of dangerous goods:
  (i) Class 1 – Explosives,
  (ii) Class 2.1 – Flammable gases, or
  (iii) Class 3 – Flammable Liquids;

Crew must have Transport Canada equivalency certificate SR 12761.1 carried while handling.

(f) **Marshalling of Autonomous Track Inspection car (ATI)**

These are cars equipped with diesel powered generator and fuel tank which must not be marshalled next to:

(i) Class 1 – Explosives,

(ii) Class 2.1 – Flammable gases, or

(iii) Class 3 – Flammable Liquids

Crew must have Transport Canada equivalency certificate SR 13140 carried while handling.

(g) **Marshalling of placarded dangerous goods cars next to unoccupied distributed power (DP) locomotives.**

A placarded car may be placed next to:

- an unoccupied engine tender,
- an unoccupied distributed power locomotive or;
- an unoccupied dead-in-tow distributed power locomotive

Provided they will not be marshalled directly next to the following:

i Class 1.1 and 1.2 – (Explosive)

ii Class 2.3 – (Toxic Gases)

iii Class 6.1 Packing Group 1 – (Toxic Inhalation)

iv Class 7 – (Radioactive) (UN2919, UN3328, UN3329, UN3331)

Crews must have Transport Canada equivalency certificate SR 13611 carried while handling.

Section 7.0 Emergency Measures

(a) A dangerous goods car discovered leaking must not be moved without authority and be kept away from switch heaters, engines, occupied passenger equipment, or any car, container, or trailer with an operating mechanical heating/ cooling device.

(b) When it has been determined that dangerous goods are involved in an incident, the following procedures must be followed:

- i Protect the movement and notify the RTC, Transportation Supervisor, or Yard Coordinator where applicable;
- ii Keep clear of the incident scene and when possible, remain up wind of cars suspected of containing dangerous goods;
- iii Take immediate action to warn other employees and, if necessary, the public;
- iv Avoid any unnecessary exposure to smoke or fumes and keep all open flames and smoking material away from the incident;
- v Determine as quickly as possible what cars are directly involved in the incident, as well as those in close proximity to them;
- vi Identify commodities involved from information contained in shipping documents;
- vii If the locomotive consist is not directly involved, and it is safe to do so, the movement should be cut as close as possible to the incident location and the remaining cars moved a safe distance;
- viii Provide the RTC, Transportation Supervisor or Yard Coordinator all pertinent information for the cars of dangerous goods involved in the incident such as:
  - car number(s);
  - contents;
  - 24 hr. emergency phone numbers;
  - condition of cars e.g. leaking, on fire etc.
- ix The RTC, Transportation Supervisor or Yard Coordinator will inform crews of emergency action to be taken to minimize the effect of the dangerous goods involved.
- x Documentation accompanying the cars involved must remain at the scene and be made available to emergency response authorities. Employees must retain possession of the documentation until relieved of that responsibility by a railway officer. **When documentation is on the RSED, a crew member must make the information available to the emergency response authorities. Crew will unlock and locate the information and accompany the device while the emergency response authorities views the information. It is permissible to transfer information to another Android device via Bluetooth.**

  While operating in the US:

  Upon request for shipping papers or emergency response information from a railroad employee, regulatory endorsement officer, or emergency response personnel in an emergency, crews will:

  - i Immediately share any requested information from the shipping papers for the shipment. Providing an extra copy of the train list/consist, when available;

    Note: retain any waybills and a copy of the train list / consist until you can deliver them to the first Railway Supervisor on the scene
  - ii Immediately provide a copy of the North American Emergency Response Guidebook

xi Employees must not speculate on the cause and should only provide pertinent information.

Section 8.0 Special Dangerous Commodities

(a) Movements carrying one or more loaded Special Dangerous Commodities will be governed by Timetable Subdivision Footnotes under the heading “Special Dangerous Commodities”.

(b) Special Dangerous Commodities will be identified on consists by the follower line “SPD - Special Commodity - 35 MPH in Canada”. The shipping document may not indicate whether the car contains a “Special Dangerous Commodity”.

When a placarded car is lifted enroute, and a consist or other means is not available to indicate that the car is not a “Special Dangerous Commodity”, such car must be handled applying “Special Dangerous Commodity” speed restrictions and inspection instructions until such time as it can be verified otherwise.

(c) Conductors will ensure that other members of the train crew are aware that Special Dangerous Commodities are part of the movement.

Conductors will also ensure that the RTC is aware that Special Dangerous Commodities are being handled in their movement:

i upon departure from the station where their train was ordered,

ii when Special Dangerous Commodities are picked up enroute.

(d) When an inspection is required or when performed in lieu of a speed restriction in compliance with the Timetable Subdivision Footnotes pertaining to Special Dangerous Commodities, it must be performed within one mile of the mileage indicated in timetable footnotes.

i A movement may be inspected by a hotbox or dragging equipment detector. When such inspection is not performed, an inspection of the equipment affected must be done at a speed not exceeding 5 MPH by one of the following:

- Mechanical department car inspectors; or
- Crews of standing trains or transfer movements; or
- Pull-by inspection by crew members; or
- Walking inspection while the movement is stopped.

ii Such inspection may be confined to the portion of the train or transfer movement from the front, back to and including the second car behind the last special dangerous commodity car. The Rail Traffic Control Centre must be notified if speed is to be reduced or the movement is to be stopped for inspection in compliance with these instructions.

(e) In the event of the failure of a Hot Box or Dragging Equipment Detector which protects a populated area (census metropolitan area), the Rail Traffic Control Centre will provide instructions with regards to the inspection requirements for that movement.

Section 9.0 KEY TRAINS and HIGHER RISK KEY TRAINS

(a) **Definition**

KEY TRAIN:

Train carrying any one or combination of the following:

- One (1) or more tank car loads of Poison or Toxic Inhalation Hazard (PIH or TIH) (Hazard Zone A, B, C, or D), anhydrous ammonia (UN1005), or ammonia solutions (UN3318);
- One (1) or more car loads of spent nuclear fuel (SNF) or high level radioactive waste (HLRW);
- A combination of twenty (20) or more car loads or intermodal portable tank loads of any combination of dangerous goods

HIGHER RISK KEY TRAIN:

Higher Risk Key Train is defined as a train carrying any combination of loaded tank cars of liquefied petroleum gases (LPG) or crude oil identified as dangerous goods and handled as following:

- 20 or more in a continuous block or;
- 35 or more throughout the train.

All applicable dangerous goods are :

"Liquefied Petroleum Gases" includes the following:	"Crude Oil" includes the following:
UN 1011 Butane	UN 1267 Petroleum Crude Oil
UN 1012 Butylene	UN 3494 Petroleum Sour Crude Oil, Flammable, Toxic
UN 1055 Isobutylene	UN 1268 *** only STCC 4906068 UN1268 Petroleum Distillates, N.O.S. (CONTAINS CRUDE OIL)
UN 1075 Liquefied Petroleum Gas	
UN 1077 Propylene	
UN 1969 Isobutane	
UN 1978 Propane	

(b) **Identifying Key Trains and Higher Risk Key Trains:**

- The WOPRT (train journal) will identify status by displaying a banner in the header block of the first page;
  - KEY TRAIN for Key Train
  - HR KEY TRN for Higher Risk Key Train
- When a WOPRT (train journal) is not available, or dangerous goods cars are picked up or set out, the conductor must review the shipping papers for all dangerous goods cars and determine Key Train or Higher Risk Key train status. If the status changes, the RTC must be promptly notified.

(c) **Instructions for operating Key Trains and Higher Risk Key Trains:**

- Maximum Speeds for Key trains and Higher Risk Key Trains:

  Key Train and Higher Risk Key Train Zones are listed **in Timetable footnote under special zone**

  The table below contains the Higher Risk Key Train speed restriction requirements.

Refer to timetable footnotes item 8 SPEEDS for the summarized Cold weather speed restrictions.

Key Train			
Signalled track		Non-Signalled Track	
	Special Zone		Special Zone
50 mph	35 mph	50 mph	35 mph

<table>
<tr><th colspan="6">Higher Risk Key Train</th></tr>
<tr><th colspan="5">Applicable on all subdivision</th><th>Subdivision specific</th></tr>
<tr><th>Temperature</th><th colspan="2">Signalled Track</th><th colspan="2">Non-Signalled Track</th><th></th></tr>
<tr><td></td><td></td><td>Special Zone</td><td></td><td>Special Zone</td><td></td></tr>
<tr><td>Warmer than -15C</td><td>50 mph</td><td>30 mph</td><td>50 mph</td><td>30 mph</td><td></td></tr>
<tr><td>- 15C to -24C</td><td>50 mph</td><td>30 mph</td><td>25 mph</td><td>25 mph</td><td>On non-signalled track, be governed by additional Subdivision restrictions as listed in special instructions which apply at -15C or colder</td></tr>
<tr><td>-25C to -29C</td><td>50 mph</td><td>25 mph</td><td>25 mph</td><td>25 mph</td><td>On Signalled track, be governed by additional Subdivision restrictions as listed in special instructions which apply at -25C or colder</td></tr>
<tr><td>-30C or colder</td><td>40 mph</td><td>25 mph</td><td>25 mph</td><td>25 mph</td><td>Be governed by the additional Subdivision restrictions as listed in special instructions.</td></tr>
<tr><td colspan="6">The following speed restrictions apply from Nov 15th to Mar 14th on subdivisions specified in special instructions as “Higher Risk Key Trains governed by date specific restrictions” below.</td></tr>
<tr><td colspan="3"><u>Signalled Track</u>
• 40 mph when temperature is -24 or warmer
• 30 mph when temperature is -25 or colder
• 25 mph when within a special zone</td><td colspan="3"><u>Non-Signalled Track</u>
Restricted to 25 mph; and be governed by additional Subdivision restrictions as listed in special instructions which will apply at -25C (non-date specific)</td></tr>
</table>

- Key Trains and Higher Risk Key Train must hold the main track at meeting or passing points when the maximum speed in the siding is 10 mph or less. Exception: Key train / Higher Risk Key Train may operate on any siding if:

  a) the non-Key/Higher Risk Key Train is a passenger train;

  b) two Key/Higher Risk Key Trains are meeting or passing;

  c) the non- Key/Higher Risk Key Train exceeds siding length;

  d) there is insufficient clearance in the siding for the non- Key/Higher Risk Key Train;

  e) the main track is impassible;

  f) the Key/Higher Risk Key Train is being staged; or

  g) the crew operating the Key/Higher Risk Key Train is going to be relieved because they have reached their regulated on duty time limit.

- Cars not equipped with roller bearings are restricted from operating in a Key train and Higher Risk Key Train.

- If a defective bearing (hot box) is identified by a WIS detector and confirmed by crew inspection, the car must be set out or, if required, the train may operate at a safe speed not exceeding 15 MPH until the car with the defective bearing is set out. If a defective bearing (hot box) is identified by a WIS detector and nothing is found during crew inspection, the movement must not exceed 30 MPH until that identified rail car has received an inspection over the next working WIS detector. If a defect in a bearing of the same car is reported by two consecutive WIS detectors, the car must be set out or must operate the Key Train or Higher Risk Key Train at a safe speed not exceeding 15 MPH until the car with the defective bearing is set out.
- When operated with Trip Optimizer, if not already programmed, the “Max Train Speed:“ must be set on the Train Setup Screen as shown in the TO Guide. The Train Setup Screen is available at Trip Initialization and also en-route. At Trip Initialization the “Max Train Speed:” will show “n/a” or may contain a value for trains that contain speed restricted cars. For Key Trains this value must be set at 50 mph (or lower, if a lower speed restriction also applies to the train or its cars)
- Unless relieved of the requirement to do so by the operating railroad’s Rail Traffic Controller, the crew operating a speed restricted Key Train and Higher Risk Key Train on a foreign railroad must, at the earliest opportunity, notify the foreign railroad’s Rail Traffic Controller that the train is a speed restricted Key Train and Higher Risk Key Train as defined by the operating railroad.

(d) **Key Routes**

When operating on “Key Routes” identified in special instructions, Key Trains and Higher Risk Key Trains must not proceed more than 40 miles without having received an inspection on both sides of the train by:

- Hot box component of a Wayside Inspection system
- Roll-by by wayside employees
- Roll-by by crews of standing trains or transfer movements
- Roll-by or walking inspection by crew member

When required to use an inspection other than a WIS inspection to comply with the 40 miles restriction, crews must notate the location and results of the inspection on the bottom of the Summary page of the WOPRT.

CN Key routes are the following:

Subdivisions	Limits
ABERDEEN	Warman to North Battleford
ALBREDA	Entire
ALLANWATER	Entire
ASHCROFT	Entire
BALA	Doncaster to Capreol
BLACKFOOT	Entire
BRAZEAU	Alix Jct to Joffre
BULKLEY	Entire
CAMROSE	Entire
CARAMAT	Entire
CHETWYND	Entire
CLEARWATER	Entire

Subdivisions	Limits
DRUMMONDVILLE	West Jct to Ste Rosalie
DUNDAS	Entire
EDSON	Entire
FORT FRANCES	Fort Frances to Rainy River
FORT ST-JOHN	Entire
FRASER	Entire
GRIMSBY	Clifton to Hamilton
HAGERSVILLE	Entire
HALTON	Entire
KINGSTON	Dorval to Pickering Jct
MONTMAGNY	St-Andre Jct to West Jct
MONTREAL	Pointe St-Charles to Dorval
NAPADOGAN	Entire
NECHAKO	Entire
NEW WESTMINSTER	Entire
OAKVILLE	Burlington West to Hamilton
PELLETIER	Entire
RAINY	Entire (See Fort Frances sub)
REDDITT	Entire
RIVERS	Entire
ROBSON	Entire
RUEL	Entire
SKEENA	Entire
SPRAGUE	Entire
SPRINGHILL	Marsh Jct to Pacific Jct
ST HYACINTHE	Ste-Rosalie to Pont Victoria
ST LAURENT	Riviere des Prairies to Josy
STAMFORD	Entire
STRATHROY	Entire
TELKWA	Entire
TETE JAUNE	Entire
THREE HILLS	Alix Jct to Mirror
VEGREVILLE	Entire
WAINRIGHT	Entire
WARMAN	Entire

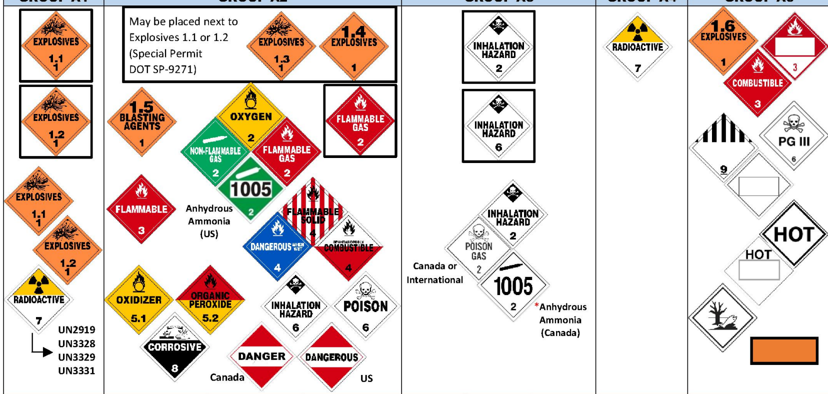
Subdivisions	Limits
WATROUS	Entire
YORK	Entire
YALE	Entire

Section 10.0 U.S. Position in Train chart

10.0 U.S. Position in train chart

Compliant with SP 20996 Rev. 2

GROUP A1	GROUP A2	GROUP A3	GROUP A4	GROUP A5



HOW TO USE THIS CHART

Select the applicable column of the Position-in-Train chart. To do so:

1. Identify the placards and/or markings applied to the car.
2. Use the shipping document to determine whether car is loaded or residue/empty. Note: The notation: "RESIDUE - LAST CONTAINED" on the shipping document indicates a residue/empty shipment.
3. Identify the car type involved by observation (e.g. tank car, hopper car, gondola, etc.).
4. Find the applicable section on the chart, based on the placard or marking applied, the load or residue/empty status, and the car type.
5. Follow the restrictions associated with the placard or marking as the "X"s in the columns indicate.

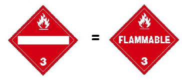
*Authorized only for U.S. to Canada or Canada to U.S. shipments.

The Word "TOXIC" can be used in place of the word "POISON" on placards.

**For instructions on marshalling Air Cars and ATIP, see General Information

Cars with placards displaying 4-digit identification numbers will be handled the same as cars with word description placards.

EQUIVALENT PLACARDS



GROUP A1	GROUP A2			GROUP A3			GROUP A4	GROUP A5		
Any Car	Loaded Tank Car	Residue Tank Car	Other Loaded Car	Loaded Tank Car	Residue Tank Car	Other Loaded Car	Any Car	Any Car		RESTRICTIONS
X	X			X						1) Must not be nearer than the 6th car from the head end engine consist (operating or dead-in-tow), occupied helper unit, or occupied caboose/business car. If the train does not have at least five buffer cars, then all available buffer cars must be placed between the placarded car and the head end engine consist (operating or dead-in-tow), occupied helper unit, or occupied caboose/business car. The available buffer cars must be equally divided to protect both head end engine consist, occupied helper unit or occupied caboose/business car. Exception: In a loaded or empty bulk commodity unit train, only one buffer car is required to be placed between the placarded car and the head end engine consist and occupied helper unit.
X				X					MUST NOT BE NEXT TO	2) Engine used as distributed power (DP) or dead-in-tow.
X	X	X		X	X		X			3) Occupied engine (head end engine consist or helper unit) or occupied caboose/business car.
X	X			X						4) Open top cars (including bulkhead flats), when any of the contents protrude beyond the car ends or, if shifted, would protrude beyond the car ends.
X	X			X						5) Loaded flat cars, except closed TOFC/COFC equipment, multi-levels, and other specially equipped cars with tie down devices for handling vehicles. Railroad wheels loaded on wheel car flats, in gondolas with no ends, or loaded with the axles above the top of the car.
X	X			X						6) Any rail cars, transport vehicles, or freight containers with temperature control equipment or internal combustion engine ** whether running or not. Note: Does not apply to cryogenic refrigerated equipment.
X	X		X	X		X	X			7) Any placarded car in another placarding Group, except it may be next to any residue placarded car or any car placarded or marked as a Group E.